Exercise 1 • Foundational Accounting Principles

# Accrual vs Cash Basis, Linear Depreciation & Inventory Sensitivity

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*A specific company has started activity on the 1st of January of the present year. To this date (31st of March), it has the following goods that were acquired on 31st of December of the previous year:*

- *Factory building: 1.000 m.u. (Duration 25 years)*
- *Office building: 1.500 m.u. (Duration 50 years)*
- *Machinery: 300 m.u. (Duration 10 years)*
- *Transport equipment: 100 m.u. (Duration 5 years)*

*Information corresponding to the 1st quarter is as follows: personnel salaries of the quarter 450 m.u. (75 m.u. pending after 31/3), materials purchases and used in production 700 m.u. (125 m.u. pending after 31/3), energy consumption estimated in 230 m.u. (nothing paid till date), sales invoiced to customers 1.930 m.u. (315 m.u. pending after 31/3). In manufacturing all sold units, the entire purchased materials have been consumed. Ignore corporation taxes.*

***a)*** *Calculate at 31/3: income, expenses, collections/cash inflows, payments/cash outflows, investments and depreciation/amortization of the quarter (assume linear depreciation of the goods along their lifetime).*

***b)*** *How will the response change if only half of the purchased materials had been consumed?*

EXAM DAY STRATEGY: ACCRUAL VS CASH & QUARTERLY PRO-RATA ALIGNMENT

- **Horizon & Pro-Rata Time:** 1st Quarter ($t = 3$ months $= \frac{1}{4}$ year $= 0.25$). Goods bought on 31/12 of the previous year $\implies$ exactly one quarter of depreciation has accrued on 31/3. Straight-line formula: $D_{\text{quarter}} = \frac{\text{Acquisition Cost}}{4 \times n}$.
- **Accrual Principle (Devengo) vs Cash Flow (Tesorería):**
  - **Revenues (Ingresos):** Recognized upon commercial delivery and invoice generation ($1,930$ mu), independent of payment terms.
  - **Collections (Cobros):** Physical cash collected $= 1,930 - 315 = \mathbf{1,615}$ mu. The pending $315$ mu becomes an asset (Accounts Receivable) on the Balance Sheet.
  - **Expenses (Gastos):** Consumed productive factors and non-cash asset wear: Salaries ($450$) + Materials consumed ($700$) + Energy consumed ($230$) + Depreciation ($30$) $= \mathbf{1,410}$ mu.
  - **Payments (Pagos):** Actual treasury disbursements $= (450 - 75) + (700 - 125) + 0 = 375 + 575 = \mathbf{950}$ mu.
- **Exam Trap 1 – Depreciation Cash Flow:** Depreciation ($30$ mu) is an accounting charge for physical/technological wear and tear. It entails **zero cash payment (**$0$ mu**)**. Never add depreciation to cash outflows!
- **Exam Trap 2 – Capital Investments (Capex):** The $2,900$ mu of fixed assets were purchased on 31/12 prior year. During the 1st quarter itself, no additional fixed assets were acquired $\implies$ Q1 Capex $= \mathbf{0}$ mu.
- **Exam Trap 3 – Inventory Sensitivity (Part b):** If only 50% of materials are consumed ($350$ mu), the remaining $350$ mu is stored as an **Inventory Asset**. Operating expenses drop by $350$ mu, but **supplier payments remain strictly** $575$ mu because payments depend on purchases invoiced, not physical consumption!

## Phase 1: Quarterly Linear Depreciation Computation

The time elapsed is the 1st Quarter ($t = 3$ months $= \frac{1}{4}$ year). The straight-line quarterly depreciation allocates asset cost evenly over its lifespan:

$$D_{\text{quarter}} = \frac{\text{Annual Depreciation}}{4} = \frac{\text{Acquisition Cost}}{4 \times n}$$

- **Factory building:** $\frac{1,000}{4 \times 25} = \frac{1,000}{100} = \mathbf{10.0}$ m.u.
- **Office building:** $\frac{1,500}{4 \times 50} = \frac{1,500}{200} = \mathbf{7.5}$ m.u.
- **Machinery:** $\frac{300}{4 \times 10} = \frac{300}{40} = \mathbf{7.5}$ m.u.
- **Transport equipment:** $\frac{100}{4 \times 5} = \frac{100}{20} = \mathbf{5.0}$ m.u.
- **Total Depreciation for the Quarter:** $10.0 + 7.5 + 7.5 + 5.0 = \mathbf{30.0}$ m.u.

## Phase 2: Resolution of Part (a) — 100% Consumed

1. **Income / Revenues (Ingresos):** Under the accrual principle, revenue is recognized upon commercial invoicing regardless of collection date:

$$\text{Income} = \text{Total Sales Invoiced} = \mathbf{1,930.0}\ \text{m.u.}$$

2. **Expenses (Gastos):** Value of all factor inputs consumed in operations during the quarter:

$$\begin{aligned} \text{Salaries Expense} &= 450.0\ \text{m.u.} \\ \text{Materials Consumed} &= 700.0\ \text{m.u.} \quad (100\%\ \text{consumed in production}) \\ \text{Energy Consumed} &= 230.0\ \text{m.u.} \\ \text{Depreciation} &= 30.0\ \text{m.u.} \\ \hline \text{Total Expenses} &= 450.0 + 700.0 + 230.0 + 30.0 = \mathbf{1,410.0}\ \text{m.u.} \end{aligned}$$

*(Quarterly Net Profit* $= \text{Income} - \text{Expenses} = 1,930.0 - 1,410.0 = \mathbf{520.0}$ m.u.*).*

3. **Collections / Cash Inflows (Cobros):** Actual liquid treasury received into the bank till:

$$\text{Collections} = \text{Total Invoiced} - \text{Pending at } 31/3 = 1,930.0 - 315.0 = \mathbf{1,615.0}\ \text{m.u.}$$

*(The uncollected* $315.0$ m.u. *is a Current Asset: Accounts Receivable).*

4. **Payments / Cash Outflows (Pagos):** Physical disbursements from company treasury:

$$\begin{aligned} \text{Salaries Paid} &= 450.0 - 75.0 = 375.0\ \text{m.u.} \quad (\text{with 75 mu Salaries Payable}) \\ \text{Materials Paid} &= 700.0 - 125.0 = 575.0\ \text{m.u.} \quad (\text{with 125 mu Accounts Payable}) \\ \text{Energy Paid} &= 230.0 - 230.0 = 0.0\ \text{m.u.} \quad (\text{entirely pending liability}) \\ \text{Depreciation Paid} &= 0.0\ \text{m.u.} \quad (\text{non-cash accounting charge}) \\ \hline \text{Total Payments} &= 375.0 + 575.0 + 0.0 + 0.0 = \mathbf{950.0}\ \text{m.u.} \end{aligned}$$

*(Net Cash Generated in Quarter* $= \text{Collections} - \text{Payments} = 1,615.0 - 950.0 = +\mathbf{665.0}$ m.u.*).*

5. **Investments (Inversiones):** Initial productive capacity was capitalized on 31/12 of the previous year ( $1,000 + 1,500 + 300 + 100 = 2,900.0$ m.u.). During Q1, no new fixed capital acquisitions occurred:

$$\text{Q1 Fixed Capital Investments} = \mathbf{0.0}\ \text{m.u.}$$

*Alternative reading:* the statement does not say whether "investments of the quarter" means new purchases or the capital invested in the goods listed. If it is read as the productive capital capitalised on 31/12 and in use during the quarter, the answer is $1,000 + 1,500 + 300 + 100 = 2,900$ m.u.

6. **Depreciation / Amortization:** Straight-line quarterly asset wear $=$ **30.0** m.u.

## Phase 3: Resolution of Part (b) — Inventory Sensitivity (50% Consumed)

If only half of the purchased materials are consumed in production ($700.0/2 = 350.0$ m.u.), the remaining 350.0 m.u. remains untouched in the warehouse as a balance sheet asset:

- **Operating Expenses:** Materials expense drops from 700 mu to 350 mu:

$$\text{New Expenses} = 450.0 + \mathbf{350.0} + 230.0 + 30.0 = \mathbf{1,060.0}\ \text{m.u.} \quad (\Delta\text{Expenses} = -350.0\ \text{m.u.})$$

$$\text{New Net Income} = 1,930.0 - 1,060.0 = \mathbf{870.0}\ \text{m.u.} \quad (\text{increased by } +350.0\ \text{m.u.})$$

- **Payments / Cash Outflows: Completely unchanged at** 950.0 m.u.**!** Supplier payments are governed by raw materials purchased and credit terms ($700 - 125 = 575$ m.u.), regardless of physical shop-floor consumption.
- **Incomes and Collections:** Unchanged at $1,930.0$ m.u. and $1,615.0$ m.u. respectively.
- **Balance Sheet Asset Impact:** Warehouse Ending Inventory increases by $+350.0$ m.u. under Current Assets.

| Financial Magnitude | Part (a): 100% Consumed | Part (b): 50% Consumed | Variation $\Delta$ |
|---|---|---|---|
| Income (Revenues) | 1,930.0 m.u. | 1,930.0 m.u. | 0 |
| Operating Expenses | 1,410.0 m.u. | 1,060.0 m.u. | -350.0 m.u. |
| **Net Income (Profit)** | **520.0 m.u.** | **870.0 m.u.** | **+350.0 m.u.** |
| Collections (Cash Inflows) | 1,615.0 m.u. | 1,615.0 m.u. | 0 |
| Payments (Cash Outflows) | 950.0 m.u. | 950.0 m.u. | 0 |
| **Net Treasury Cash Flow** | **+665.0 m.u.** | **+665.0 m.u.** | **0** |
| Ending Inventory Asset | 0.0 m.u. | 350.0 m.u. | +350.0 m.u. |
| Quarterly Depreciation | 30.0 m.u. | 30.0 m.u. | 0 |

PEDAGOGICAL CONCLUSION & GOLDEN RULE

Unconsumed purchased materials represent a **Balance Sheet Current Asset (Inventory)**, not an Income Statement expense. While Net Income increases by 350 m.u., treasury cash flow remains identical. **Profit $\neq$ Cash Flow!**

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Study notes written from the official course material (T3_Financial Statements_EXERCISES.pdf; Topic 3 Financial Management (I).pdf). Online version: subjects/business-management/problemas/topic-3-financial-statements.html